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CFD analysis of water gas shift membrane reactor

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ABSTRACT

Fuel cell based modular power generation can be achieved by miniaturization and process intensification of equipments in the process. Fuel cells require hydrogen rich gas which can be generated through reforming and water gas shift reaction. The water gas shift reactor being kinetically limited occupies more volume to achieve the required CO conversion. A membrane reactor integrates the reaction and hydrogen separation stages and hence reduces the volume requirement. Computational Fluid Dynamics offers virtual prototyping of the reactor and thus helps in design, optimization and scale up of reactors. In this study customized User Defined Functions (UDFs) were developed to analyze the performance of low temperature water gas shift membrane reactor. The models were validated using literature data for the parameters – synthesis gas compositions, time factor, sweep flow rate and steam to CO ratio. The effect of all these parameters on the reactor was analyzed for CO conversion, H_2 recovery, DaPe, concentration polarization, concentration profiles and conversion index. The simulations have showed that the UDFs developed were capable of simulating the membrane reactor and this can be used for the design and optimization of the membrane reactor for any process conditions.

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Keywords: Membrane reactor; CFD; Palladium membrane; Water gas shift reaction

1. Introduction

The technological advancements towards hydrogen economy are driving the development of portable power systems like remote community power generators and onboard power generators for transportation sector. In these power generators, the technologies involved are reforming of fuel followed by water gas shift reactor, gas separation and fuel cells. Among the technologies mentioned above, the water gas shift reaction which reduces the CO content and enriches the H_2 content in the synthesis gas is kinetically limited and hence occupies the maximum volume (Francesconi et al., 2007). Hence it is imperative that this process component be modified to suit modularity. One process intensification step that helps in this route is the membrane reactor technology whereby both the reaction and separation can be achieved in a single process vessel. The concept of membrane reactor applied to the water gas shift reactor thus will have the advantages of reduced size of vessel, increased conversion of CO (due to continuous removal of product the equilibrium shifts to the right) and effective separation of H_2 in a single vessel.

In the tubular membrane reactor configuration, the water gas shift catalyst is packed in the interior reactor and the walls of the reactor is replaced by a hydrogen permeable membrane and the diffusing hydrogen is tapped through the annular space. Even though hydrogen can be permeated by both organic and inorganic membranes, inorganic membranes are suitable owing to their mechanical strength and temperature tolerance. The most studied membrane for water gas shift reaction is Palladium and its alloys with Silver and Copper due to their higher selectivity and permeability to hydrogen.

Several experimental studies on use of membrane reactors for water gas shift reaction have been carried out using different catalysts and membrane materials by Kikuchi et al. (1989), Uemiya et al. (1991), Basile et al. (1996a,b, 2001, 2003, 2008), Chiappetta et al. (2006a,b), Barbieri et al. (2008) and Brunetti et al. (2009). Criscuoli et al. (2000) compared the performance of Pd, and mesoporous reactors of similar volume on three different gas feed mixtures and found the Pd membrane reactor reporting up to 100% conversion at higher time factors and 595 K. The high temperature operation of water gas shift reaction was carried out by Iyoha et al. (2007a,b) in multi

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tubular membrane reactor with Pd and Pd/Cu membranes in the absence of catalysts. Mendes et al. (2010) have recently made a review of the water gas shift reaction in membrane reactors.

Since carrying out experiments to study the feasibility, optimization of operating parameters and scale up of membrane reactors is very costly and time consuming, modeling and simulation of the membrane reactor using existing data provides an easy methodology. Several attempts have been made to model the membrane reactors till date. The performance of the membrane reactor has been characterized by H_2 recovery index, conversion index and volume index (Brunetti et al., 2009). Since the water gas shift reaction is a catalytic reaction, both homogeneous and heterogeneous models have been used to model the reactor. Some of the notable models reported are by Damle et al. (1994), Alfadhel and Kothare (2005a,b), Barbieri et al. (2005), Brunetti et al. (2007), Chiappetta et al. (2008) and Adrover et al. (2009, 2010). Even though several models exist, many cater to the specific experimental data and general membrane reactor models are few. Solution strategies are difficult to implement and the models seldom predict the partial pressure variations close to the membrane which has a prominent role in the driving force for membrane permeation. Computational Fluid Dynamics (CFD) can be used for virtual prototyping of chemical reactors and since it is based on control volume methodology, the local variations of the fluid, thermal and mass transport properties can be visualized in comparison to simple models and can be used to design the reactors. Since permeation is based on local conditions near to the membrane face, CFD is the best tool to analyze membrane reactors. Very few CFD studies have been carried out on the water gas shift membrane reactor using different gas mixtures. Coroneo et al. (2009a,b) have used the commercial CFD solver FLUENT to analyze the effect of operating conditions on Pd/Ag and ceramic membrane modules.

2. Geometry and boundary conditions

The experimental results of Criscuoli et al. (2000) are taken as the basis for validation of the CFD codes as this is a study carried out with different compositions of feed gases and operating conditions. The geometry of the membrane reactor used in the simulation is made of two concentric tubes namely the lumen and shell. The lumen is made of the membrane material and is packed with catalyst and this region acts as the reacting side. The membrane zone has an inner diameter of 8 mm and length of 150 mm representing a total reactor volume of 7.5 cm^3. The shell has an inner diameter of 40 mm with provision for sweep gas inlet and the overall length of the reactor is 280 mm. The catalyst particles were assumed to be spherical particles having a constant diameter of 1 mm each. Further, porosity of the bed of 0.457 has been imposed in the simulations and the above values have been arrived based on the physical characteristics provided by Criscuoli et al. (2000). The characteristics of the catalyst used by Criscuoli et al. (2000) represent Topsoe make (LK-821-2) with the following properties; composition $CuO–ZnO–Al_2O_3$, density 2400 ($kg\,m^{-3}$), specific heat 687 ($J\,kg^{-1}\,K^{-1}$), thermal conductivity 1.163 ($W\,m^{-1}\,K^{-1}$), porosity 0.457, permeability 2.158×10^{-9} (m^2), inertial resistance 19,912 (m^{-1}) and surface area/volume 3256 ($m^2\,m^{-3}$). The hydrogen permeation through the membrane wall is modeled using the permeation equation

$$Q_{H_2} = 2.95 \times 10^{-4} \exp\left(\frac{-5833.5}{T}\right) \ (\mathrm{mol\,m\,m^{-2}\,s^{-1}\,Pa^{-0.5}})$$

where it is dependent on the trans-membrane hydrogen partial pressure adjacent to the membrane. The membrane is assumed to be of zero thickness as the resistance to hydrogen permeation through the alumina support is assumed negligible and the thickness of Palladium membrane is 70–75 μm with infinite selectivity for hydrogen.

The 3D geometry of the reactor described above was generated using the pre-processor GAMBIT 2.3 and solved using FLUENT 6.3, a commercial CFD code based on finite volume methodology using Cartesian co-ordinate system. The geometry was discretized using unstructured grids, with number of nodes varying as 198,828, 284,130, 418,232 and 753,559. Boundary layers were generated on either side of the membrane with the first row having a thickness of 0.1 mm. One cell thick zones were segregated on either side of the membrane to apply the permeation of hydrogen. Custom designed User Defined Functions (UDFs) have been developed to impose the volumetric reaction, source terms and momentum terms into the solver. Simulations using these grid sizes were compared for flow parameters and CO conversion and the grid size corresponding to 284,130 nodes provided sufficiently accurate results without further improvement with increase in number of cells and hence are used for the simulation. The reacting zone is treated as a porous zone and the physical velocity condition is imposed in the simulation. The viscous loss term and the inertial loss terms were computed in comparison with Ergun's equation and used in the simulation. To check the pressure drop and velocity of the reactor, flow simulations were carried out without the reactions and the results were compared with the Ergun's equation and manual computation of velocity. The cold flow velocity profile and pressure drop computed by the simulation were similar with that values computed manually showing that the formulation of the flow field is correct.

The simulations have been carried out assuming that the flow represents a three dimensional, steady state, incompressible, laminar flow with isothermal conditions. Further, the gas has been assumed to obey ideal gas equation. The incompressible flow has been imposed in the simulations as the maximum velocity of the gas inside the packed bed reactor represents Mach number $\ll 0.3$, indicating density variations are negligible. The binary diffusion coefficients used in computations have been computed using the Fullers equation (Cussler, 1998) and fitted as a second order polynomial equation. The reaction is modeled using pseudo homogeneous reaction approach since the porosity of the bed was constant owing to smaller regular particle size and the extent of reaction was considered uniform all along the length of the reactor. Thus volumetric reaction rate was sufficient to model the reaction kinetics. The solutions have been obtained using segregated solver approach with second order upwind scheme for discretization. The SIMPLE (Semi Implicit Pressure Linked Equation) algorithm was used for the pressure velocity coupling with pressure term discretized using the PRESTO (Pressure Staggering Option) scheme, whose predictions were found to be better for porous bed. The convergence criterion was fixed as 1E-05 for momentum equations and 1E-07 for all other equations. The CFD simulations were carried out for three gas mixtures of different compositions designated as Mix 1, Mix 2, and Mix 3. These mixtures repre-

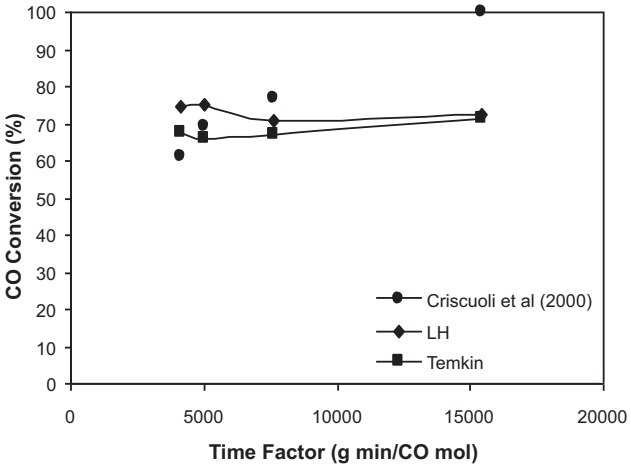


Fig. 1 – Simulated results for Mix 1, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and $Q_{sweep} = 43.6$ ml min^{-1}.

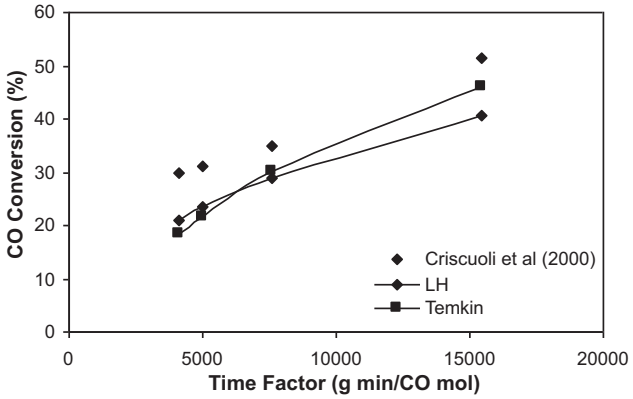


Fig. 2 – Simulated results for Mix 2, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and $Q_{sweep} = 43.6$ ml min^{-1}.

sent the compositions expected in gasification and reformed synthesis gases. Mix 1 and Mix 2 are typical compositions found in gasification. Their compositions on dry basis are

Mix 1: CO (32%), CO_2 (12%), H_2 (4%) and N_2 (52%);
Mix 2: CO (12.27%), CO_2 (11.49%), H_2 (75%) and CH_4 (1.24%);
Mix 3: CO (27%) and H_2 (73%).

3. Results and discussion

The CFD code developed and implemented in FLUENT was validated against the experimental data reported in the literature for all the operational parameters for a membrane reactor. The important operational parameters that affect the performance of membrane reactors are temperature, pressure, time factor, sweep flow rate and steam to CO ratio. Since experimental studies by Criscuoli et al. (2000) have found 595 K as the optimum operating temperature, all the simulations were carried out at that temperature. The effects of rest of the parameters are analyzed using CO conversion and H_2 recovery. The CO conversion is calculated using the inlet and exit CO flow rates and H_2 recovery is defined as the ratio of H_2 flow rate in permeate to H_2 flow rate in retentate and permeate. Kinetic screening studies on packed bed reactor by the authors have found the Langmuir Hinshelwood model and Temkin model given in appendix better predicting the reaction and hence have been employed in this study.

3.1. *Time factor*

Time factor corresponds to the ratio of catalyst loading to the reactant gas flow rate and thus is an indicator of the residence time of the reactor. The simulations were carried out at time factors of 4140, 5000, 7610 and 15,450 g_{cat} min CO mol^{-1} under 595 K, 1 atm pressure at lumen and shell, $H_2O/CO = 1.1$ and Nitrogen sweep flow rate of 43.6 ml min^{-1}. Figs. 1–3 represent the results of the simulation and they are compared against experimental values available at different conditions. The CO conversion increases with time factor and experimentally it reaches 100% for the time factor of 15,450 for Mix 1. But the CFD simulations could reach only 71% using both the kinetic expressions for the reaction. For Mix 2, the variations in CO conversion could be easily captured by both the kinetic expressions and Temkin model is better than the Langmuir Hinshelwood model in predicting the conversion. This is a departure from the fixed bed reactor where the results by Langmuir Hinshelwood model were better. For Mix 3 too the conversion of Temkin model was higher than that by Langmuir Hinshelwood model. Fig. 3 shows the comparison of Mix 3 with sweep flow of 436 ml min^{-1}. Here too the Temkin model is able to predict the conversion better. The maximum conversion of 90% reported could be captured by the simulation with a corresponding H_2 recovery of 93%. The CFD simulations show that the Temkin model is comparatively better than the Langmuir Hinshelwood model in predicting the water gas shift reaction in a membrane reactor. Hence the rest of the simulations were carried out with the Temkin model for the reaction.

3.2. *Sweep flow*

Since the permeation of H_2 across the membrane is based on the trans-membrane partial pressure of H_2, one way of increasing the conversion and H_2 recovery is by using an inert gas in the shell as sweep gas flow as co-current flow. The flow rate of this sweep gas (N_2) in the shell side has a profound effect on H_2 permeation and hence the conversion in the reactor. To analyze the influence of the sweep gas flow rate, three sets of simulations were carried out namely, without sweep flow and with sweep flow of 43.6 ml min^{-1} and 436 ml min^{-1} at different time factors and gas mixture compositions. The positive impact of sweep flow could be very well seen in Figs. 4–6. In all the three gas mixtures, the conversion increased across

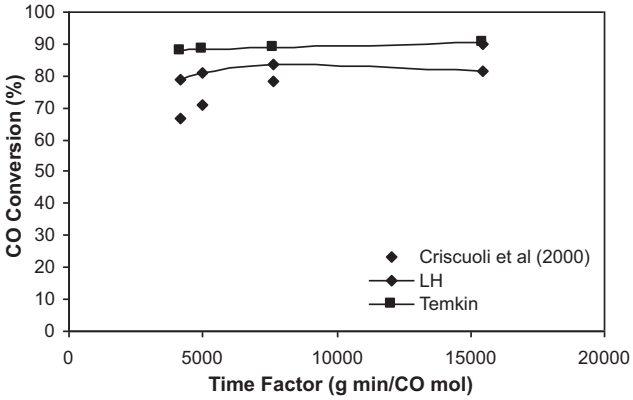


Fig. 3 – Simulated results for Mix 3, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and $Q_{sweep} = 436$ ml min^{-1}.

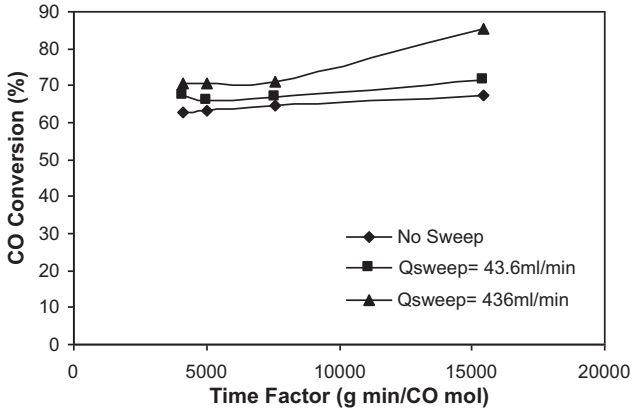


Fig. 4 – Effect of sweep flow rate on Mix 1, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm.

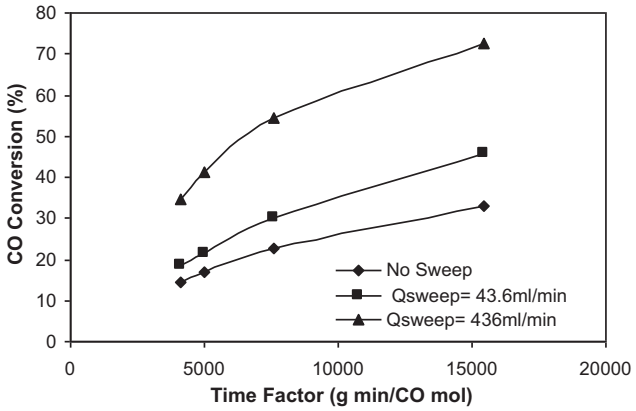


Fig. 5 – Effect of sweep flow rate on Mix 2, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm.

the time factors. Since Mix 1 was thermodynamically favorable (has the highest equilibrium conversion) for more conversion, the effect of sweep gas flow was not much significant. In both Mix 2 and Mix 3 which were less thermodynamically active, the increase in sweep flow rate 10 times significantly increased the conversion. The maximum conversions reported at the highest time factor under study increased to 72% for Mix 2 and 90% for Mix 3.

The hydrogen recovery too is greatly impacted by the sweep flow rate. The H_2 recovery was insignificant in the simulation without sweep flow and increased with sweep flow. The H_2

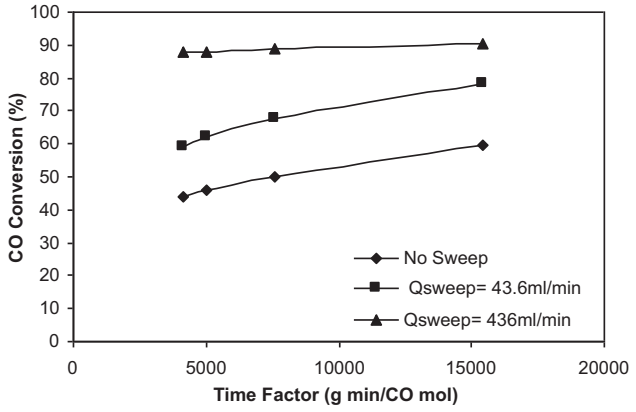


Fig. 6 – Effect of sweep flow rate on Mix 3, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm.

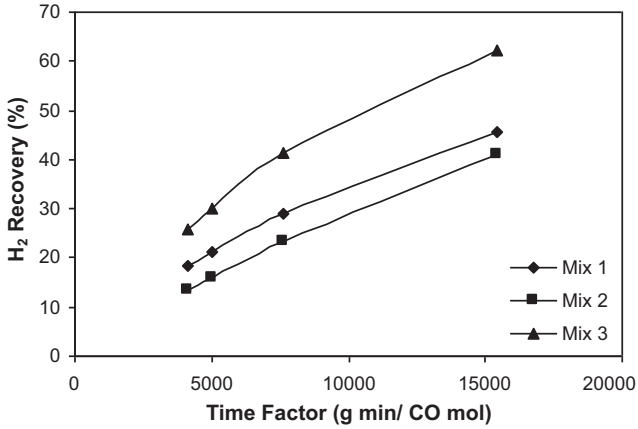


Fig. 7 – Hydrogen recovery in the membrane reactor for $Q_{sweep} = 43.6\ ml\ min^{-1}$, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm.

recoveries for both sweep flows are depicted in Figs. 7 and 8. All the three mixtures showed increase in H_2 recovery with increase in sweep flow rate. Among the three mixtures, Mix 3 has reported the highest H_2 recoveries with both the sweep flow rates.

3.3. *Steam to CO ratio*

Excess steam is always fed to the reactor and the ratio of steam to CO is another parameter that determines the conversion in the reactor. To identify the effect of steam to CO ratio on a membrane reactor, simulations were carried out by keeping the temperature, pressure and sweep gas flow constant and varying the feed gas mixture as per the various steam to CO ratios. The conditions maintained are 595 K, $Q_{sweep} = 436\ ml\ min^{-1}$, $P_{lumen} = P_{shell} = 1$ atm and Temkin kinetic model for reaction. The results obtained have been plotted in Fig. 9. The introduction of excess steam increased the conversion of the reaction in all the three mixtures. The simulations indicate an optimum steam to CO ratio of 2–2.5 for all the three mixtures. Fig. 10 is the effect of steam to CO ratio on the H_2 recovery from the membrane reactor. Interestingly the H_2 recovery decreases for all the three mixtures with increase in the steam to CO ratio. This shows the inhibiting nature of excess steam on the permeation through the membrane.

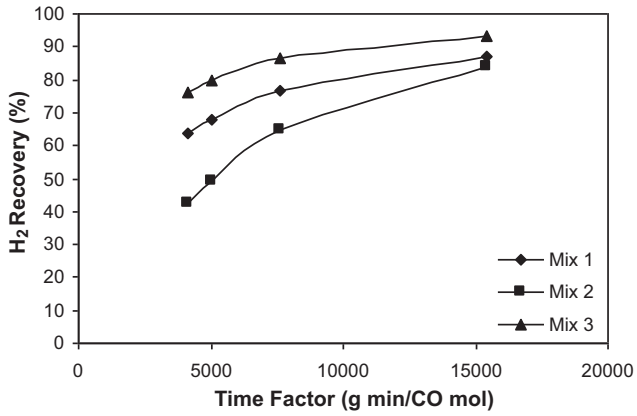


Fig. 8 – Hydrogen recovery in the membrane reactor for $Q_{sweep} = 436\ ml\ min^{-1}$, 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm.

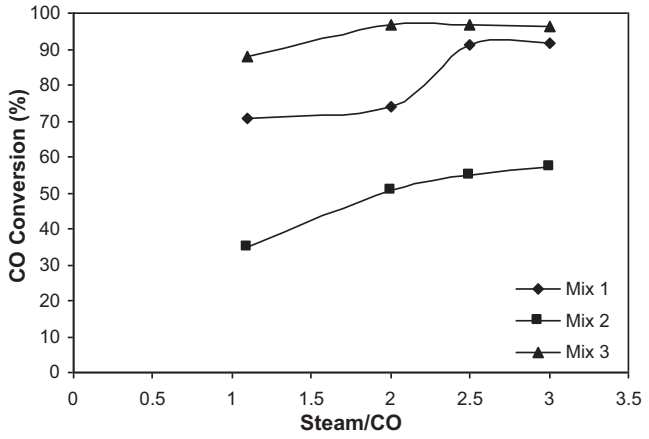


Fig. 9 – Effect of steam to CO ratio on the membrane reactor for Q_{sweep} = 436 ml min^{-1}, 595 K and $P_{lumen} = P_{shell}$ = 1 atm.

3.4. *Concentration profiles*

Study of the concentration profiles inside the reactor helps us in understanding the reaction rate occurring as well as the quantity of catalyst and hence the length requirement of the reactor. In a water gas shift membrane reactor, the important reactant to be traced is CO as the conversion is dependent on this gas and is the rate limiting component. On the shell side the gas of importance to us is hydrogen and tracking the concentration of H_2 gives an idea on the membrane requirement for the membrane reactor. Due to space constraints, the CO concentrations for time factor 4140 g min CO mol^{-1} alone of the three gas mixtures are provided in Figs. 11–13 For Mix 1 at time factors of 4140 and 5000 g min CO mol^{-1}, the CO concentration decreases exponentially along the dimensionless length of the reactor till 80% and then the concentration remains constant. This shows that the reaction is active till 80% of the length of the reactor. For higher time factors, this length reduces to 70% of the length of the reactor. But for the highest time factor and highest sweep flow rate, the concentration keeps decreasing till the full length of the reactor. In accordance with the higher conversion with the highest sweep gas, the CO profile for Mix 2 has the largest variation for the highest sweep gas. It is interesting to note that for all the four time factors studied, the concentration profiles showed a reduction in concentration till 60% of the reactor length and beyond that the concentration variation is very small. This

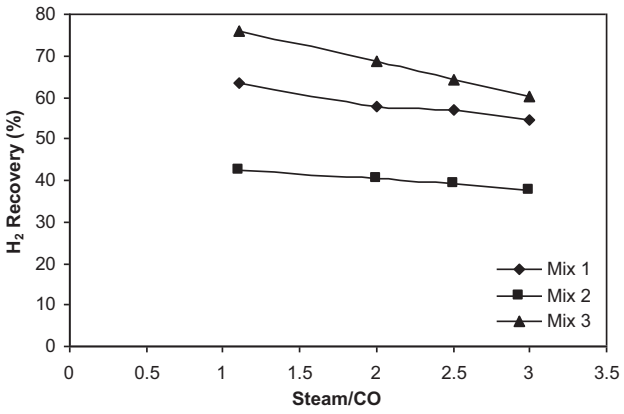


Fig. 10 – Hydrogen recovery for varying steam to CO ratio on the membrane reactor for Q_{sweep} = 436 ml min^{-1}, 595 K and $P_{lumen} = P_{shell}$ = 1 atm.

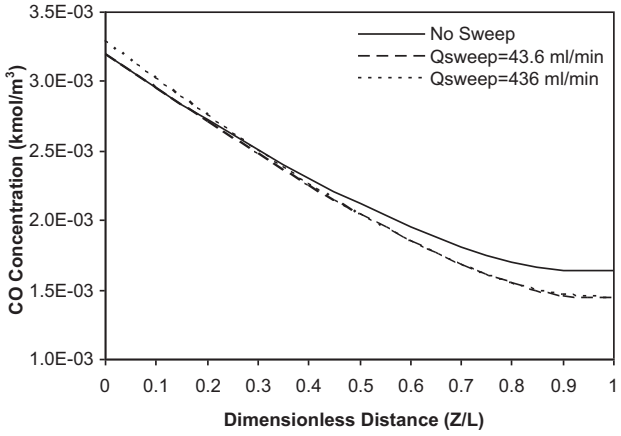


Fig. 11 – Lumen CO profile along the length of the reactor for Mix 1 and time factor = 4140 g min CO mol^{-1}.

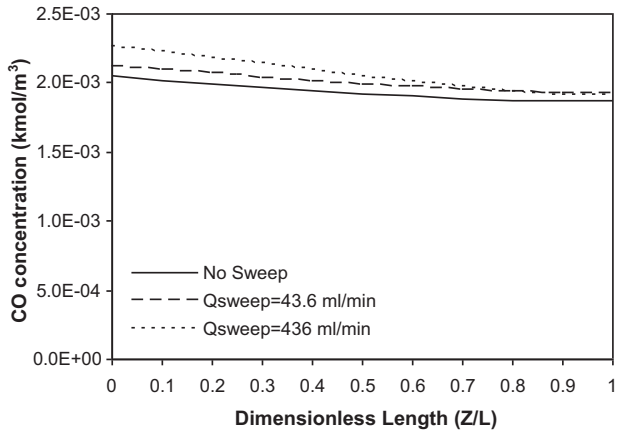


Fig. 12 – Lumen CO profile along the length of the reactor for Mix 2 and time factor = 4140 g min CO mol^{-1}.

indicates that for carrying out the water gas shift reaction for Mix 2, 60% of the existing volume is sufficient to achieve the maximum possible conversion. This observation applies to all the three conditions of operation namely, no sweep flow, sweep flow = 43.6 ml min^{-1} and sweep flow = 436 ml min^{-1}. For Mix 3 the order of CO reduction is no sweep gas followed by sweep flow = 43.6 ml min^{-1} and 436 ml min^{-1}. The drop in CO concentration is highest with sweep flow of 436 ml min^{-1} and the percentage change is higher for time factors of 4140 and 5000 g min CO mol^{-1} respectively. The analysis of the length

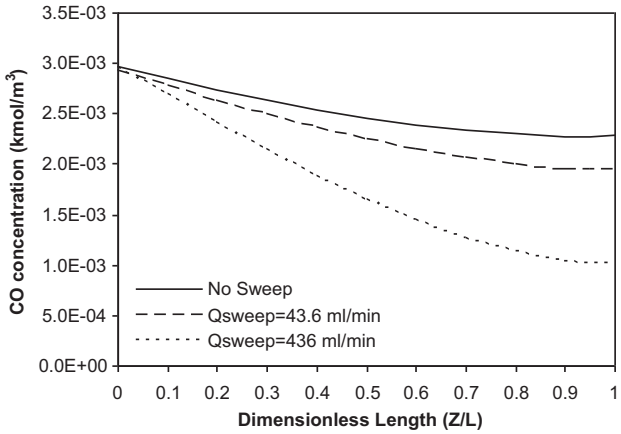


Fig. 13 – Lumen CO profile along the length of the reactor for Mix 3 and time factor = 4140 g min CO mol^{-1}.

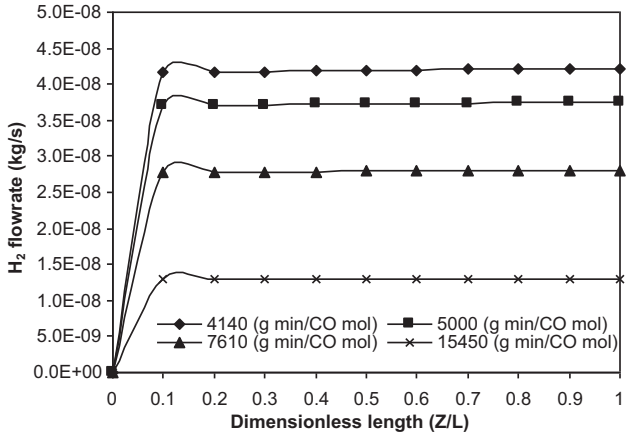


Fig. 14 – Shell side H_2 profile along the length of the reactor for Mix 1 and Q_{sweep} = 436 ml min^{-1}.

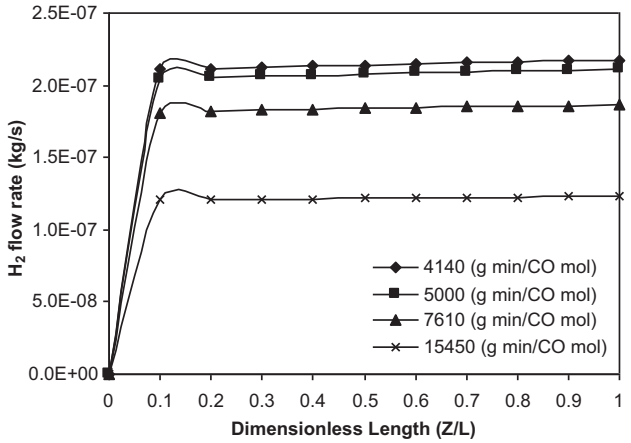


Fig. 15 – Shell side H_2 profile along the length of the reactor for Mix 2 and Q_{sweep} = 436 ml min^{-1}.

requirement for the reaction shows that for the time factors of 4140 and 5000 g min CO mol^{-1}, 80% of the length of the reactor is sufficient. For higher time factors a further reduction of 10% in length requirement is reported. The H_2 profile along the shell for all the three mixtures is reported for the sweep flow rates of 43.6 ml min^{-1} and 436 ml min^{-1} respectively in Figs. 14–16. In all these figures it is observed that the highest quantity of H_2 permeated lies within the first 20% of the length of the shell and then slightly increases along the length of the shell.

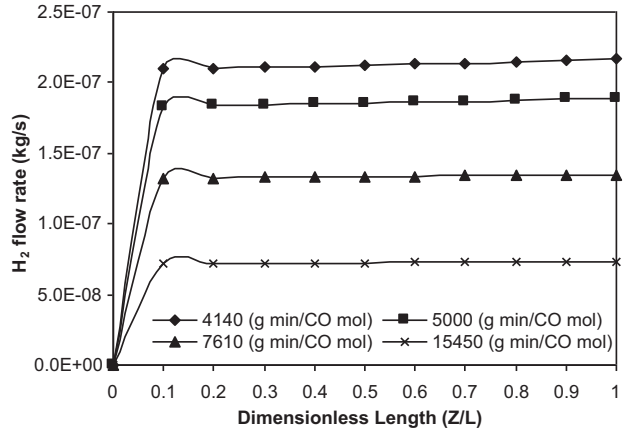


Fig. 16 – Shell side H_2 profile along the length of the reactor for Mix 3 and Q_{sweep} = 436 ml min^{-1}.

3.5. *Analysis of DaPe*

An important design factor put forth for membrane reactors is the product of the dimensionless numbers Damkohler Number (Da) and Peclet Number (Pe). When DaPe equals to 1, then all of the H_2 that is generated by reaction is removed through the membrane. This sets the limit i.e. the maximum permeation possible to match the reaction rate. If DaPe < 1, it signifies that the reactor capacity is not being fully utilized. If DaPe > 1, then H_2 builds up in the reactor and productivity is constrained by an inadequate membrane. For the three different mixtures analyzed in this study, the dimensionless numbers Da and Pe were computed at different time factors. The typical DaPe value for Mix 1 was 0.0323 which is very much lower than 1, which suggests that the specified membrane area available in the reactor is in excess of the requirement for this gas mixture. The same result has been obtained by tracking the concentration profile for Mix 1. The DaPe for Mix 2 is found to be 1.5815 which suggests the inability of the available membrane area to effectively transport H_2. It should be noted here that Mix 2 has high H_2 content in the feed and the reaction rate is very small. This leads to more H_2 accumulating in the retentate side of the membrane reactor. This result too could be seen in the concentration profiles of CO and H_2 in Mix 2. The DaPe values for Mix 3 which showed the maximum benefit of using the membrane reactor were found to be 0.6995 which is pretty close to the optimum value of 1. So this analysis using the dimensionless numbers DaPe shows that the available catalyst and the available membrane area in the membrane reactor under study was insufficient for carrying out the reaction mixture Mix 2, where as it was greatly in excess of requirement for Mix 1 and slightly in excess for Mix 3.

3.6. *Analysis of concentration polarization*

Concentration polarization (CP) is a phenomenon observed in membrane separation processes where due to the selective permeation of one component, the other components accumulate near the membrane surface thereby creating a resistance to the radial flow of the component. Thus concentration polarization reduces the flux of the permeating component and is a major design parameter in membrane processes. Since various equations are available to calculate concentration polarization, the hydrogen polarization number proposed by Jacopo et al. (2009) for Pd/Ag membrane is used in this analysis. The hydrogen concentration polarization number (S) proposed is a dimensionless quantity which indicates the relative importance of the mass transfer resistances offered by the gas phase and by the metal membrane respectively. For a gas permeable metal membrane system, $S \gg 1$ indicates that the membrane permeance is so high that the overall resistance is controlled by concentration polarization in the external gas phase while on the contrary, when $S \ll 1$ the permeance of the metal membrane is so small that it controls the entire process resistance. In all the cases under consideration, the concentration polarization number for this membrane reactor is very much less than 1 as depicted in Figs. 17–19. The effect of time factor did not change the concentration polarization number much under no sweep flow and sweep flow = 43.6 ml min^{-1}. The simulation without any sweep gas flow in the shell showed the lowest concentration polarization number which shows that the membrane diffusion is very slow without any sweep flow. For Mix 1 whose CP

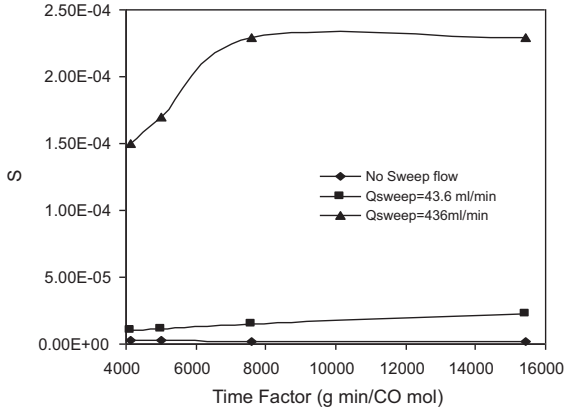


Fig. 17 – Hydrogen concentration polarization at different sweep flow rates for Mix 1.

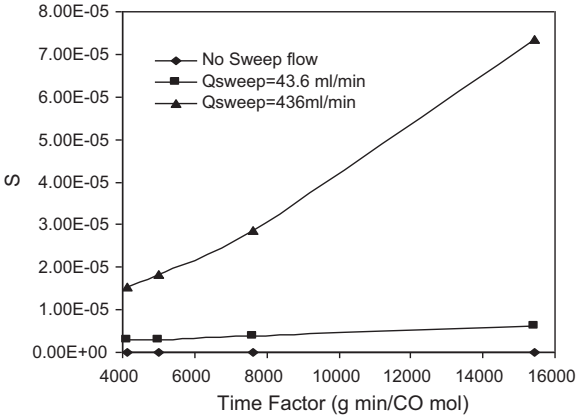


Fig. 18 – Hydrogen concentration polarization at different sweep flow rates for Mix 2.

is the highest among the three mixtures, the influence of feed flow rate was not much significant as can be seen in Fig. 17. Considerable increase in CP could be seen with the highest sweep flow rate in the shell indicating that the permeance of hydrogen is faster with high sweep flow rate. For Mix 2 and Mix 3, the CP increases linearly with time factor and higher sweep flow rate. This effect is attributed to the composition of Mix 2 and Mix 3 which has the highest fraction of H_2 in the gas mixtures and Mix 1 had very less H_2 in the mixture. Thus

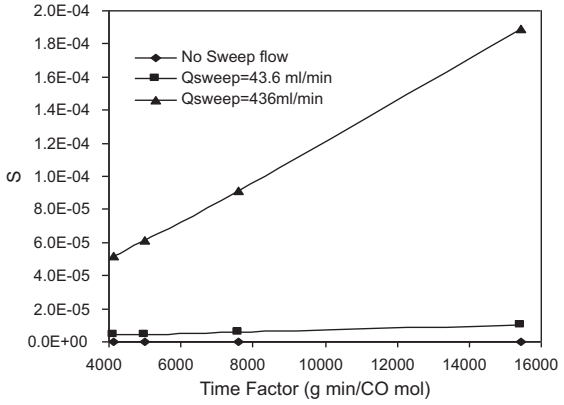


Fig. 19 – Hydrogen concentration polarization at different sweep flow rates for Mix 3.

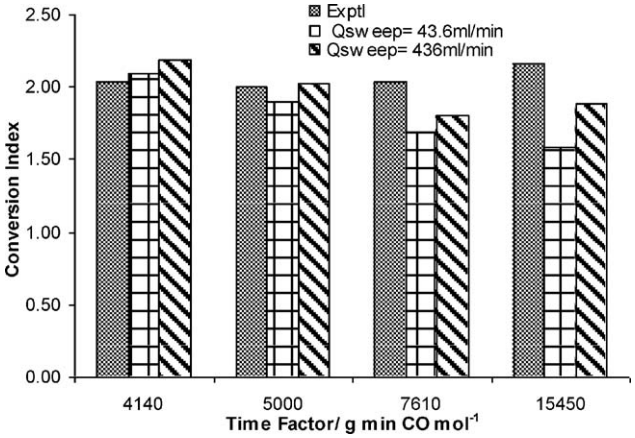


Fig. 20 – Conversion index for Mix 1 at 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and experimental value with $Q_{sweep} = 43.6\ ml\ min^{-1}$.

it is proved that in a membrane reactor, the role of concentration polarization is negligible and does not need to be included in the design of Palladium membrane reactors for water gas shift reaction. This analysis shows that the permeation in the Pd membrane is limiting the process and not the mass transfer considerations.

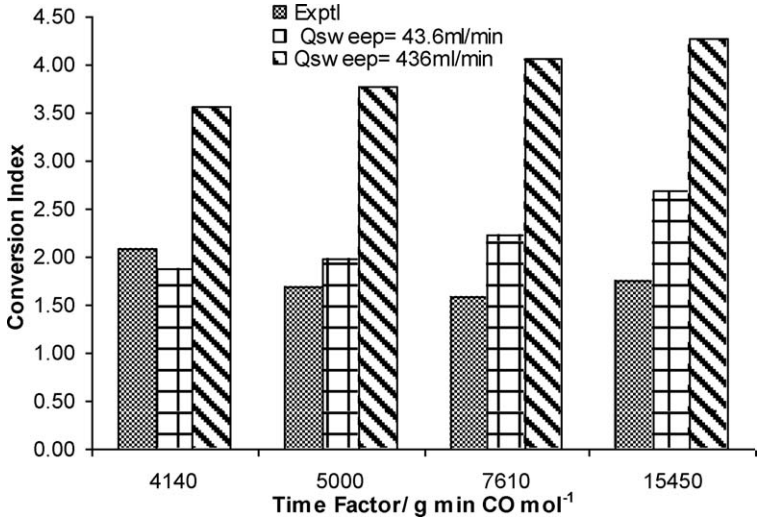


Fig. 21 – Conversion index for Mix 2 at 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and experimental value with $Q_{sweep} = 43.6\ ml\ min^{-1}$.

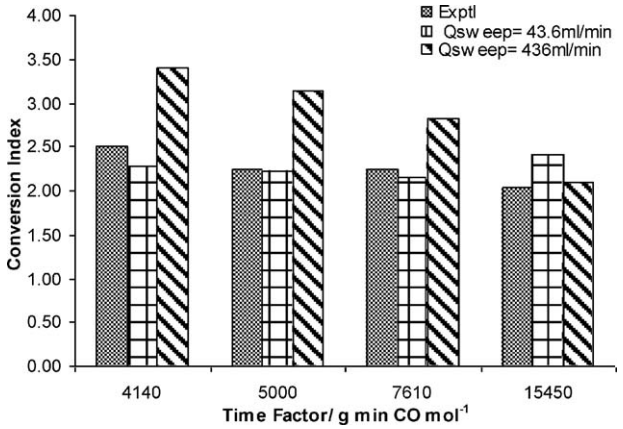


Fig. 22 – Conversion index for Mix 3 at 595 K, $H_2O/CO = 1.1$, $P_{lumen} = P_{shell} = 1$ atm and experimental value with $Q_{sweep} = 436\ ml\ min^{-1}$.

3.7. *Conversion index*

Conversion index is the ratio of the CO conversion in a membrane reactor to the CO conversion in a traditional reactor at the same reactor volume. It thus is a measure of the comparative increase in conversion that can be achieved by the membrane reactor and also the volume requirement for carrying out the reaction. The conversion index was calculated for the gas mixtures at the experiments with $Q_{sweep} = 43.6\ ml\ min^{-1}$ and simulated results with $Q_{sweep} = 43.6$ and $436\ ml\ min^{-1}$ and are reported in Figs. 20–22. The average conversion index for Mix 1 is around 2 which shows that the membrane reactors can double the conversion for the same volume of reactor. The influence of membrane reactor on a gas mixture with all the gases present i.e. Mix 2 is the highest. This mixture is thermodynamically less active and hence using a membrane reactor can increase the conversion approximately four times than that of the traditional reactor. For Mix 3 also the conversion is 2.5 times more. This shows that the use of membrane reactor has a profound effect on the volume of the reactor required.

Thus it can be seen that the 3D simulation carried out using CFD could be used to extensively analyze the various operating conditions in a membrane reactor and can visualize and understand the membrane reactor operation and performance better than the simple models. Moreover the UDFs developed are general in nature that, this can be used to simulate any membrane reactor provided the kinetics of the reaction and the permeability of the membrane under operation is known.

4. Conclusion

The water gas shift converter in a modular power generation system can be reduced in size by the process intensification of the reactor and H_2 separator stages into a single unit. The membrane reactor thus used poses engineering challenges in its design and scaleup as both reaction step and permeation step has to be incorporated in the same reactor. Computational Fluid Dynamics offers an option of virtual walk through the reactor. CFD studies offer the local fluctuations of the operating parameters and hence are better in designing the membrane reactors. The metallic membrane reactor of annular configuration was studied using the commercial CFD code FLUENT by varying the operating parameters of synthesis gas composition, time factor, sweep gas flow rate and steam to CO ratios at 595 K. The reaction kinetics that was better for predicting the water gas shift reaction under membrane reactor conditions was the Temkin model. The studies show that the CO conversion in all the cases of membrane reactors were considerably higher than the traditional reactor and occupied less space. The introduction of sweep gas in the shell side also increased the CO conversion because of the increase in trans-membrane partial pressure difference. The simulations showed an optimum steam to CO ratio of 2–2.5 for the reaction. The conversion index for all the 3 mixtures showed the possibility of reducing the volume requirement for the reactor. Thus CFD was capable of reproducing the experimental results in a membrane reactor and this methodology can be used to design and scale up membrane reactors for any process conditions expected in a modular power generator.

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Appendix A.

A.1. *Langmuir Hinshelwood model*

$$r = k K_{CO} K_{H_2O} \left[P_{CO} P_{H_2O} - \left(\frac{P_{CO_2} P_{H_2}}{K_{eq}} \right) \right] (1 + K_{CO} P_{CO} + K_{H_2O} P_{H_2O} + K_{CO_2} P_{CO_2})^{-2} \left(\frac{\rho_{cat}}{60} \right) (\text{mol cm}^{-3}\ \text{s}^{-1})$$

$$K_{eq} = \exp\left(\frac{4577.8}{T} - 4.33 \right)$$

$$k = \exp\left(-\frac{29367}{1.987 \times T} + \frac{40.32}{1.987} \right) \text{mol}\, g_{cat}^{-1}\, \text{min}^{-1}$$

$$K_{CO} = \exp\left(\frac{3064}{1.987} - \frac{6.74}{1.987} \right) \text{atm}^{-1}$$

$$K_{H_2O} = \exp\left(-\frac{6216}{1.987 \times T} + \frac{12.77}{1.987} \right) \text{atm}^{-1}$$

$$K_{CO_2} = \exp\left(\frac{12542}{1.987 \times T} - \frac{18.45}{1.987} \right) \text{atm}^{-1}$$

P_i is partial pressure (atm), i corresponds to the species.

A.2. *Temkin model*

$$r = k \left[P_{CO} P_{H_2O} - \left(\frac{P_{CO_2} P_{H_2}}{K_{eq}} \right) \right] (A P_{H_2O} + P_{CO_2})^{-1} (\text{s}^{-1})$$

$$K_{eq} = \exp\left(\frac{4577.8}{T} - 4.33 \right)$$

$$k = 6 \times 10^{11} \exp\left(-\frac{26800}{1.987 \times T} \right) \text{atm}^{-1}\, \text{s}^{-1}$$

$$A = 2.5 \times 10^{9} \exp\left(-\frac{21500}{1.987 \times T} \right)$$

P_i is partial pressure (atm), i corresponds to the species.

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.cherd.2011.02.031.

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